

Uttkarsh Thakur

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SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C++

Backend: Spring Boot, Spring Web MVC, Spring Data JPA, Hibernate, REST API Design, Bean Validation, FastAPI, Flask, Node.js

Databases: PostgreSQL, MySQL, H2, MongoDB, Flyway Migrations, JPA/Hibernate ORM, Schema Design

Frontend: React.js, TypeScript, Tailwind CSS, Vite, HTML/CSS

Testing & Tools: JUnit 5, Mockito, AssertJ, Docker, Maven, Git & GitHub, Swagger/OpenAPI, Postman

AI/ML: RAG Pipelines, Semantic Search, FAISS, LangChain, Transformers, PyTorch, TensorFlow, Scikit-learn

PROJECTS

BillSync | Java 21, Spring Boot, Spring Data JPA, PostgreSQL, Flyway, React, TypeScript, Docker 2026

- Built a **full-stack web application** for splitting group expenses: members log what they paid, and the app works out who owes whom, backed by a **Java Spring Boot** REST API of **16 endpoints** and a **React** and **TypeScript** interface.
- Solved a rounding problem that makes shared balances drift over time, so every split adds up to the **exact total** with no lost fractions; verified automatically across thousands of amount and group-size combinations.
- Developed a **settle-up feature** that turns a tangle of debts into the fewest payments possible, reducing a 5-member group with 8 outstanding debts to just **4 transfers**.
- Designed the **PostgreSQL** database of **6 tables** with version-controlled migrations, and tuned a slow screen that made **22 database calls** down to **1**.
- Wrote **73 automated tests** covering business rules and every API endpoint, so new changes ship without breaking existing behaviour.

CodeNav AI | FastAPI, React.js, LangChain, FAISS, Sentence-Transformers, OpenAI API, GitPython 2026

- Built an **AI-powered search tool** that lets a developer ask questions about an unfamiliar codebase in plain English and get back the relevant code, using **large language models** and **semantic search (RAG)**.
- Improved answer quality by splitting code along real function and class boundaries instead of fixed-size blocks, so results come back complete and in context.
- Cut the time to open a codebase from **over 20 seconds to under 0.1 seconds**, a **99.5% improvement**, by saving the search index to disk instead of rebuilding it each time.

Motion Sensing System | TensorFlow/Keras, VGG16, Flask, Streamlit, OpenCV, Python 2024

- Built a system that **recognises human activities from live video**, adapting a pre-trained image-recognition model (**VGG16**) rather than training from scratch, and delivered it as a live dashboard.

RESEARCH & TALKS

Agreement Loops in Multi-Agent Systems | Seminar Talk, Literature Review 2026

- Reviewed current research on **how independent AI agents reach agreement** with one another, and presented the findings as a **seminar talk** covering what works today and what remains unsolved.

EDUCATION

Bennett University Aug. 2023 – May 2027

B.Tech in Computer Science and Engineering | **CGPA: 8.17/10** Greater Noida, India

- Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Deep Learning, Neural Networks, System Design

DAV Public School Graduated 2022

Senior Secondary Education (Class XII) Ranchi, India

CERTIFICATIONS

Build Basic Generative Adversarial Networks (GANs) — DeepLearning.AI via Coursera Apr. 2026

Neural Networks and Deep Learning — DeepLearning.AI via Coursera Apr. 2025

Machine Learning: Clustering & Retrieval — University of Washington via Coursera Nov. 2024

ACHIEVEMENTS

Ranked **101st** nationally at **Smart India Hackathon 2024**

CodeChef: 2-star rated | **LeetCode:** Contest rating **1500+**

Competed in **Hackaccino Hackathon**, Bennett University (2024)